THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science |

Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 4 N	Balanced forces.
2	**(d)** 600 Pa	240 / 0.4 = 600.
3	**(b)** particles move faster at higher temperature	Higher kinetic energy.
4	**(d)** O_2 and Fe	Oxygen and iron.
5	**(b)** 20 per cent	40/200 x 100.
6	**(d)** positive charge	Like charges repel.
7	**(c)** more air is pumped in	More collisions.
8	**(a)** latent heat of vaporisation	Extra energy absorbed.
9	**(c)** mercury	Sulphur is the non-metal.
10	**(c)** colloid	Tyndall effect.
11	**(a)** change in colour	Force does not change colour.
12	**(b)** the same type of storm with different names	Tropical cyclones.
13	**(b)** liquid	Liquefaction.
14	**(b)** sonorous and hard	Alloy property.
15	**(c)** more concentrated	More solute.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Net = 15 - 6 = 9 N (1); In the direction of push (1).	1+1
17	P = 80/0.2 (1); = 400 Pa (1).	1+1
18	Particles of one fit into the spaces of the other (1); This shows that there are gaps between particles (1).	1+1
19	H and O are chemically combined in a fixed ratio (1); It has properties different from H2 and O2 (1).	1+1
20	Solution = 100 g (1); Percentage = 18 per cent (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	25 N is static friction at the limit (1); 18 N is sliding friction (1); Static friction is larger than sliding friction (1).	1+1+1
22	P = 400 / 0.02 = 20000 Pa (1); Two feet: area 0.04, P = 10000 Pa (1); Pressure is double on one foot (1).	1+1+1
23	Solids: strong forces hold particles (1); Liquids: particles slide over each other (1); Gases: particles fly apart (1).	1+1+1
24	Black mass forms (1); Magnet does not attract it (1); A compound is formed (1).	1+1+1
	OR: Mass of solution = 175 g (1); 35/175 x 100 (1); = 20 per cent (1).	1+1+1
25	Moon's gravity is less (1); Weight is gravitational force (1); Mass is the amount of matter (1).	1+1+1
	OR: Cyclone: large, spirals over warm sea (1); Tornado: small, funnel on land (1); Thunderstorm: lightning and rain (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Readings increase with roughness (1); Friction depends on nature of surface (1); Carpet has the highest (1); Adding mass increases normal force (1); Readings increase (1).	1+1+1+1+1
	OR: Tractor area 1.2 square metres (1); Pressure 36000/1.2 = 30000 Pa (1); Person: 600/0.05 = 12000 Pa (1); Tractor pressure is greater but wheels are wide (1); Wide tyres prevent sinking (1).	1+1+1+1+1

27	Place a crystal in water (1); Colour spreads without stirring (1); Motion of particles (1); Spaces: dissolves with little volume change (1); Faster in warm water (1).	1+1+1+1+1
	OR: Mixture: magnet separates iron (1); Mixture retains yellow and grey colours (1); After heating: black non-magnetic iron sulphide (1); New properties (1); Compound formed by chemical change (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 8 N	20 - 12 = 8 N.
2	**(c)** 500 Pa	20/0.04 = 500 Pa.
3	**(a)** particles of salt fit in the gaps between water particles	Spaces between particles.
4	**(a)** NH_3 and CH_4	Ammonia and methane.
5	**(c)** 30 g	0.12 x 250.
6	**(c)** it induces opposite charge near it	Charge separation.
7	**(c)** outside pressure is lower than inside	Lower external pressure.
8	**(a)** boiling point (condensation point)	Condensation.
9	**(d)** brass	Copper is an element.
10	**(a)** milk	Colloid.
11	**(a)** 0.3 kg	3/10.
12	**(b)** over land under thunderclouds	From severe thunderstorms.
13	**(d)** evaporates quickly	Takes heat.
14	**(b)** a good conductor and ductile	Properties.
15	**(d)** temperature and amount of sugar	Constants.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Spikes increase friction with the ground (1); Prevents slipping (1).	1+1
17	Heated air expands and becomes less dense (1); The balloon floats up (1).	1+1
18	Higher temperature increases particle speed (1); Diffusion becomes faster (1).	1+1
19	C: 2, H: 6, O: 1 (1); Total 9 atoms (1).	1+1
20	Stirring only speeds up dissolving (1); Solubility depends on temperature and nature (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Vapour: particles far apart (1); Condenses: closer, slide (1); Freezes: fixed in a pattern (1).	1+1+1
22	Lustrous (1); Malleable and ductile (1); Do not rust easily (1).	1+1+1
23	Prepare a hot saturated solution (1); Filter and cool slowly (1); Crystals form and are dried (1).	1+1+1
24	At the poles (1); Strongest magnetic force is at the poles (1); Few at the middle (1).	1+1+1
	OR: Squeezing the bulb removes air (1); Releasing it lowers the pressure inside (1); Atmospheric pressure fills it with liquid (1).	1+1+1
25	Larger area exposes more particles (1); More evaporate (1); A plate of water dries faster than a glass (1).	1+1+1
	OR: Bells: sonorous (1); Utensils: conduct heat (1); Wires: ductile and conduct current (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Variable: temperature of water (1); Same volume of water in each beaker (1); Add solute until saturated and record the mass (1); Table of masses at 20, 40, 60 degrees Celsius (1); Both rise with temperature, sugar more (1).	1+1+1+1+1
	OR: Feather: large air resistance (1); Stone: small compared with weight (1); Parachutist: opens a parachute (1); Air resistance grows with speed (1); Reaches constant speed (1).	1+1+1+1+1

27	Before: warnings, shelters, supplies (1); Before: secure loose objects (1); During: stay inside (1); After: avoid flood water and wires (1); Help and report (1).	1+1+1+1+1
	OR: Compress the gas (1); Cool it (1); LPG (1); CNG and liquid oxygen (1); Liquid nitrogen (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science |

Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 0 N	Balanced.
2	**(a)** 200 Pa	20/0.1 = 200 Pa.
3	**(a)** solid, liquid, gas	Solids have the least gaps.
4	**(b)** compound	New substance.
5	**(b)** 475 g	Salt = 25 g.
6	**(a)** charge flows to earth	Grounded.
7	**(c)** land is warmer than sea in the daytime	Air rises over land.
8	**(c)** water at 90 degrees Celsius	Higher temperature.
9	**(b)** silicon	Metalloid features.
10	**(b)** sand in water	Suspension.
11	**(d)** have two poles	Poles always occur in pairs.
12	**(d)** eye wall	Around the eye.
13	**(b)** vapour cannot escape; condensation balances evaporation	Equilibrium.
14	**(d)** good conductors of heat	Heat conduction.
15	**(a)** 4 g remains undissolved	Saturation reached.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Friction between hands (1); Energy changes to heat (1).	1+1
17	Air tries to equalise the pressure difference (1); Moves from where the pressure is more (1).	1+1
18	Vapour is the gaseous form of a substance that is normally liquid at room temperature (1); Gas is a substance in gaseous state at room temperature, e.g. oxygen (1).	1+1
19	Water has two hydrogen atoms and one oxygen atom (1); Formula is H2O (1).	1+1
20	Lemon water is a true solution with tiny particles (1); Milk particles are larger and scatter (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Evaporation (1); Distillation (1); Crystallisation (1).	1+1+1
22	Same direction: 20 + 12 = 32 N (1); Opposite directions: 20 - 12 = 8 N (1); Net force depends on direction (1).	1+1+1
23	Bottle with holes at heights (1); Lowest hole shows the farthest jet (1); Pressure increases with depth (1).	1+1+1
24	Large gaps in gases (1); Gas particles move fast and freely (1); Fewer collisions slow movement (1).	1+1+1
	OR: Magnet: attracts mixture only (1); Dilute acid: iron sulphide gives hydrogen sulphide smell (1); Mixture shows yellow and grey particles (1).	1+1+1
25	Water is 90 per cent (1); Solution = 360/0.9 = 400 g (1); Sugar = 40 g (1).	1+1+1
	OR: Pull a block with a spring balance (1); Add weights on the block (1); Reading increases (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Iron in blood (1); Calcium in bones (1); Oxygen to breathe (1); Copper wires (1); Carbon fuels (1).	1+1+1+1+1
	OR: Solubility decreases (1); Crystals separate (1); Remaining solution stays saturated (1); Mass of crystals depends on the difference (1); Used in crystal preparation (1).	1+1+1+1+1

27	(a) Static friction (1); (b) Sliding friction (1); (c) Rolling friction (1); (d) Static friction (1); Rolling is the smallest (1).	1+1+1+1+1
	OR: Pressure rises with depth (1); Acts in all directions (1); Jets from side holes (1); Dam wall is thicker at the base (1); Independent of container shape (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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